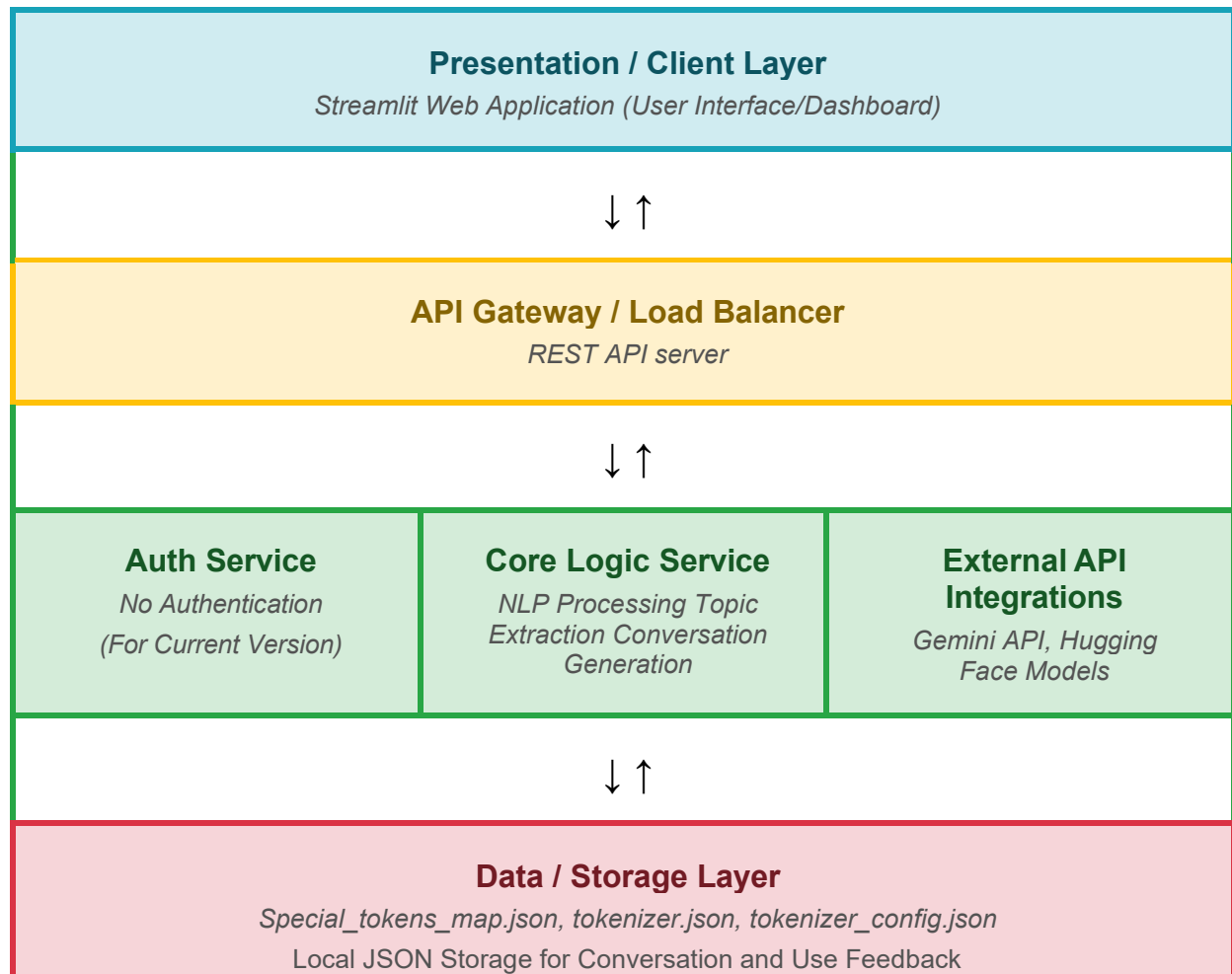


Solution Architecture

Date	02 July 2026
Team ID	xxxxxx
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides user interface for analyzing emotions, generating responses using two models (BiLSTM, BERT or Both)	Streamlit, HTML, CSS
API Gateway	Handles incoming API requests, validates data and routes requests to appropriate services	Gemini API
Microservices	Performs emotion analysis, topic extraction, conversation generation, history logging, and fact verification	Python, Transformers
Database	Stores conversation history and user emotion analysis for future references	JSON Files
External API Integration	Retrieves verified information and NLP-Based responses.	Hugging Face Transformers